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jakka_bhanu@LAPTOP-DCEV: ~$ mkdir ~/practical1
jakka_bhanu@LAPTOP-DCEVSCQ0:~$ cd ~/practical1
jakka_bhanu@LAPTOP-DCEVSCQ0:~/practical1$ nano run_command.c
jakka_bhanu@LAPTOP-DCEVSCQ0:~/practical1$ gcc run_command.c -o run_command
jakka_bhanu@LAPTOP-DCEVSCQ0:~/practical1$ ./run_command
Enter a Linux command to run: ls -l
Parent process PID: 662
Child process PID: 663 (Parent PID: 662)
total 24
-rwxr-xr-x 1 jakka_bhanu jakka_bhanu 16520 Aug 27 04:49 run_command
-rw-r--r-- 1 jakka_bhanu jakka_bhanu 1285 Aug 27 04:49 run_command.c
Child (PID 663) finished with exit code 0
Parent (PID 662) is done.
jakka_bhanu@LAPTOP-DCEVSCQ0:~/practical1$ uname -a
Linux LAPTOP-DCEVSCQ0 6.18.33.2-microsoft-standard-WSL2 #1 SMP PREEMPT_DYNAMIC Thu Jun 18 21:54:43 UTC 2026 x86_64 GNU/Linux
jakka_bhanu@LAPTOP-DCEVSCQ0:~/practical1$ lscpu
Architecture:          x86_64
CPU op-mode(s):        32-bit, 64-bit
Address sizes:          46 bits physical, 48 bits virtual
Byte Order:             Little Endian
CPU(s):                 22
On-line CPU(s) list:   0-21
Vendor ID:              GenuineIntel
Model name:             Intel(R) Core(TM) Ultra 7 155H
CPU family:             6
Model:                  170
Thread(s) per core:     2
Core(s) per socket:     11
Socket(s):              1
Stepping:               4
BogoMIPS:               5990.40
Flags:                  fpu vme de pse tsc msr pae mce cx8
                        apic sep mtrr pge mca cmov pat pse3

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BogoMIPS:              5990.40
Flags:                  fpu vme de pse tsc msr pae mce cx8
                        apic sep mtrr pge mca cmov pat pse3
                        6 clflush mmx fxsr sse sse2 ss ht s
                        yscall nx pdpe1gb rdtscp lm constan
                        t_tsc rep_good nopl xtopology tsc_r
                        eliable nonstop_tsc cpuid tsc_known
                        _freq pni pclmulqdq vmx ssse3 fma c
                        x16 pcid sse4_1 sse4_2 x2apic movbe
                        popcnt tsc_deadline_timer aes xsav
                        e avx f16c rdrand hypervisor lahf_l
                        m abm 3dnowprefetch ssbd ibrs ibpb
                        stibp ibrs_enhanced tpr_shadow ept
                        vpid ept_ad fsgsbase tsc_adjust bmi
                        1 avx2 smep bmi2 erms invpcid rdsee
                        d adx smap clflushopt clwb sha_ni x
                        saveopt xsavec xgetbv1 xsaves avx_v
                        nni vnmi umip waitpkg gfni vaes vpc
                        lmulqdq rdpid movdiri movdir64b fsr
                        m md_clear serialize ibt flush_l1d
                        arch_capabilities

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Virtualization features:

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Virtualization:        VT-x
Hypervisor vendor:     Microsoft
Virtualization type:   full

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Caches (sum of all):

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L1d:                    528 KiB (11 instances)
L1i:                    704 KiB (11 instances)
L2:                     22 MiB (11 instances)
L3:                     24 MiB (1 instance)

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NUMA:

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NUMA node(s):           1
NUMA node0 CPU(s):      0-21

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jakka_bhanu@LAPTOP-DCEV: x + v
NUMA node0 CPU(s): 0-21
Vulnerabilities:
Gather data sampling: Not affected
Ghostwrite: Not affected
Indirect target selection: Not affected
Itlb multihit: Not affected
L1tf: Not affected
Mds: Not affected
Meltdown: Not affected
Mmio stale data: Not affected
Old microcode: Not affected
Reg file data sampling: Not affected
Retbleed: Mitigation; Enhanced IBRS
Spec rstack overflow: Not affected
Spec store bypass: Mitigation; Speculative Store Bypass
                    s disabled via prctl
Spectre v1: Mitigation; usercopy/swapgs barrier
             s and __user pointer sanitization
Spectre v2: Mitigation; Enhanced / Automatic IB
             RS; IBPB conditional; PBRSE-IBRS S
             W sequence; BHI BHI_DIS_S
Srbds: Not affected
Tsa: Not affected
Tsx async abort: Not affected
Vmscape: Not affected
jakka_bhanu@LAPTOP-DCEVSCQ0:~/practical1$ lsblk
NAME MAJ:MIN RM SIZE RO TYPE MOUNTPOINTS
sda 8:0 0 356.9M 1 disk
sdb 8:16 0 159.4M 1 disk
sdc 8:32 0 2G 0 disk [SWAP]
sdd 8:48 0 1T 0 disk /mnt/wslg/distro

/
jakka_bhanu@LAPTOP-DCEV: x + v
/
jakka_bhanu@LAPTOP-DCEVSCQ0:~/practical1$ ps aux
USER PID %CPU %MEM VSZ RSS TTY STAT START TIME COMMAND
root 1 0.2 0.1 24336 15408 ? Ss 04:46 0:00 /sbin/init
root 2 0.0 0.0 3180 2204 hvc0 Sl+ 04:46 0:00 /init
root 7 0.0 0.0 3180 2044 hvc0 Sl+ 04:46 0:00 plan9 --control-socket 7 --log-level 4 --server-fd 8 --pipe-f
root 47 0.0 0.2 50400 16604 ? S<s 04:46 0:00 /usr/lib/systemd/systemd-journald
systemd+ 82 0.0 0.1 22416 14480 ? Ss 04:46 0:00 /usr/lib/systemd/systemd-resolved
root 89 0.1 0.1 35504 12484 ? Ss 04:46 0:00 /usr/lib/systemd/systemd-udev
root 127 0.0 0.0 2888 1948 ? Ss 04:46 0:00 /bin/sh /usr/lib/systemd/scripts/chronyd-starter.sh -n -F 1
root 134 0.0 0.0 4472 3168 ? Ss 04:46 0:00 /usr/sbin/cron -f -P
message+ 137 0.0 0.0 8800 5400 ? Ss 04:46 0:00 @dbus-daemon --system --address=systemd: --nofork --nopidfile
root 180 0.0 0.3 44096 29816 ? Ss 04:46 0:00 /usr/bin/python3 /usr/bin/networkd-dispatcher --run-startup-t
root 187 0.0 0.1 18436 9396 ? Ss 04:46 0:00 /usr/lib/systemd/systemd-logind
root 193 0.0 0.1 1866032 15696 ? Ssl 04:46 0:00 /usr/libexec/wsl-pro-service
syslog 219 0.0 0.0 220548 5660 ? Ssl 04:46 0:00 /usr/sbin/rsyslogd -n -iNONE
root 243 0.0 0.0 5492 2768 tty1 Ss+ 04:46 0:00 /usr/sbin/agetty --noreset --noclear --issue-file=/etc/issue:
root 302 0.0 0.4 123460 32708 ? Ssl 04:46 0:00 /usr/bin/python3 /usr/share/unattended-upgrades/unattended-up
_chrony 305 0.0 0.1 23600 11112 ? S 04:46 0:00 /usr/sbin/chronyd -n -F 1 -x
_chrony 331 0.0 0.0 12096 2456 ? S 04:46 0:00 /usr/sbin/chronyd -n -F 1 -x
root 408 0.0 0.0 3184 1108 ? Ss 04:46 0:00 /init
root 410 0.0 0.0 3200 1248 ? S 04:46 0:00 /init
jakka_b+ 412 0.0 0.0 6268 5532 pts/0 Ss 04:46 0:00 -bash
root 415 0.0 0.0 8644 5588 ? Ss 04:46 0:00 login -- jakka_bhanu
jakka_b+ 466 0.0 0.1 22344 13544 ? Ss 04:46 0:00 /usr/lib/systemd/systemd --user
jakka_b+ 470 0.0 0.0 22916 4092 ? S 04:46 0:00 (sd-pam)
jakka_b+ 498 0.0 0.0 6136 5392 pts/1 Ss+ 04:46 0:00 -bash
jakka_b+ 722 0.0 0.0 7160 4392 pts/0 R+ 04:51 0:00 ps aux
jakka_bhanu@LAPTOP-DCEVSCQ0:~/practical1$ top
top - 04:55:59 up 9 min, 1 user, load average: 0.03, 0.05, 0.02
Tasks: 27 total, 1 running, 26 sleeping, 0 stopped, 0 zombie
%Cpu(s): 0.0 us, 0.0 sy, 0.0 ni, 100.0 id, 0.0 wa, 0.0 hi, 0.0 si, 0.0 st
MiB Mem : 7692.2 total, 6348.8 free, 613.3 used, 883.5 buff/cache
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jakka_bhanu@LAPTOP-DCEV: x + v
jakka_b+ 722 0.0 0.0 7160 4392 pts/0 R+ 04:51 0:00 ps aux
jakka_bhanu@LAPTOP-DCEVSC00:~/practical1$ top
top - 04:56:17 up 9 min, 1 user, load average: 0.02, 0.05, 0.02
Tasks: 25 total, 1 running, 24 sleeping, 0 stopped, 0 zombie
%Cpu(s): 0.0 us, 0.0 sy, 0.0 ni, 100.0 id, 0.0 wa, 0.0 hi, 0.0 si, 0.0 st
MiB Mem : 7692.2 total, 6344.3 free, 617.8 used, 883.5 buff/cache
MiB Swap: 2048.0 total, 2048.0 free, 0.0 used, 7074.4 avail Mem

  PID USER      PR  NI    VIRT    RES    SHR S  %CPU  %MEM     TIME+ COMMAND
    1 root        20   0   24336   15408  11584 S   0.0   0.2   0:00.85 systemd
    2 root        20   0    3180    2204   2072 S   0.0   0.0   0:00.01 init-systemd(Ub
    7 root        20   0    3180    2044   1940 S   0.0   0.0   0:00.00 init
   47 root        19  -1   50400   16688   15400 S   0.0   0.2   0:00.18 systemd-journal
   82 systemd+    20   0   22416   14480   11984 S   0.0   0.2   0:00.09 systemd-resolve
   89 root        20   0   35504   12484   9260 S   0.0   0.2   0:00.58 systemd-udev
  127 root        20   0    2888    1948   1816 S   0.0   0.0   0:00.07 chronyd-starter
  134 root        20   0    4472    3168   2924 S   0.0   0.0   0:00.00 cron
  137 message+    20   0    8800    5404   4724 S   0.0   0.1   0:00.06 dbus-daemon
  180 root        20   0   44096   29816   14680 S   0.0   0.4   0:00.29 networkd-dispat
  187 root        20   0   18436   9396   8152 S   0.0   0.1   0:00.10 systemd-logind
  193 root        20   0  1866032   15712  12736 S   0.0   0.2   0:00.15 wsl-pro-service
  219 syslog      20   0  220548    5660   4364 S   0.0   0.1   0:00.08 rsyslogd
  243 root        20   0    5492    2768   2580 S   0.0   0.0   0:00.01agetty
  302 root        20   0  123460   32708  18092 S   0.0   0.4   0:00.22 unattended-upgr
  305 _chrony      20   0   23600   11112   7128 S   0.0   0.1   0:00.13 chronyd
  331 _chrony      20   0   12096    2456   1812 S   0.0   0.0   0:00.01 chronyd
  408 root        20   0    3184    1108    980 S   0.0   0.0   0:00.00 SessionLeader
  410 root        20   0    3200    1248   1108 S   0.0   0.0   0:00.02 Relay(412)
  412 jakka_b+    20   0    6268    5532   3864 S   0.0   0.1   0:00.07 bash
  415 root        20   0    8644    5588   4748 S   0.0   0.1   0:00.01 login
  466 jakka_b+    20   0   22344   13608  10988 S   0.0   0.2   0:00.14 systemd
  470 jakka_b+    20   0   22916   4092   2088 S   0.0   0.1   0:00.00 (sd-pam)
  498 jakka_b+    20   0    6136    5392   3848 S   0.0   0.1   0:00.03 bash
```